Measures of Central Tendency

Measures of Central Tendency are statistical tools used to find the *central* or *typical* value in a set of data. They help us summarize large amounts of data into a single representative value.

There are **three main measures**:

1️⃣ Mean (Average)

$$Mean = (sum of all values)/(number of values)$$

✅ Explanation:

The **mean** is calculated by adding all the values together and dividing by the total number of values.

📌 Example:

Marks: 5, 7, 8, 10

Mean = (5 + 7 + 8 + 10) ÷ 4
Mean = 30 ÷ 4 = **7.5**

✔ Advantages:

- Uses all values in the data
- Good for further statistical calculations

❌ Disadvantages:

- Affected by **extreme values (outliers)**

2️⃣ Median

$$Median = middlevaluewhendataisarrangedinorder$$

✅ Explanation:

The **median** is the middle number when the data is arranged from smallest to largest.

📌 Example:

Marks: 3, 6, 8, 9, 12

Median = **8** (middle value)

If there are two middle numbers:
Marks: 2, 4, 6, 8
Median = (4 + 6) ÷ 2 = **5**

✔ Advantages:

- Not affected by extreme values
- Suitable for skewed data

❌ Disadvantages:

- Does not use all values fully

3️⃣ Mode

$$Mode = valuethatoccursmostfrequently$$

✅ Explanation:

The **mode** is the number that appears most often in a data set.

📌 Example:

Marks: 4, 6, 6, 7, 9

Mode = **6**

✔ Advantages:

- Easy to find
- Useful for categorical data (e.g., favorite color)

❌ Disadvantages:

- There may be more than one mode (bimodal or multimodal)
- Sometimes no mode exists

📘 Comparison Table

Measure	Best Used When	Affected by Outliers?
Mean	Data is normally distributed	Yes
Median	Data is skewed	No
Mode	Categorical or repeated data	No